Computer Networks

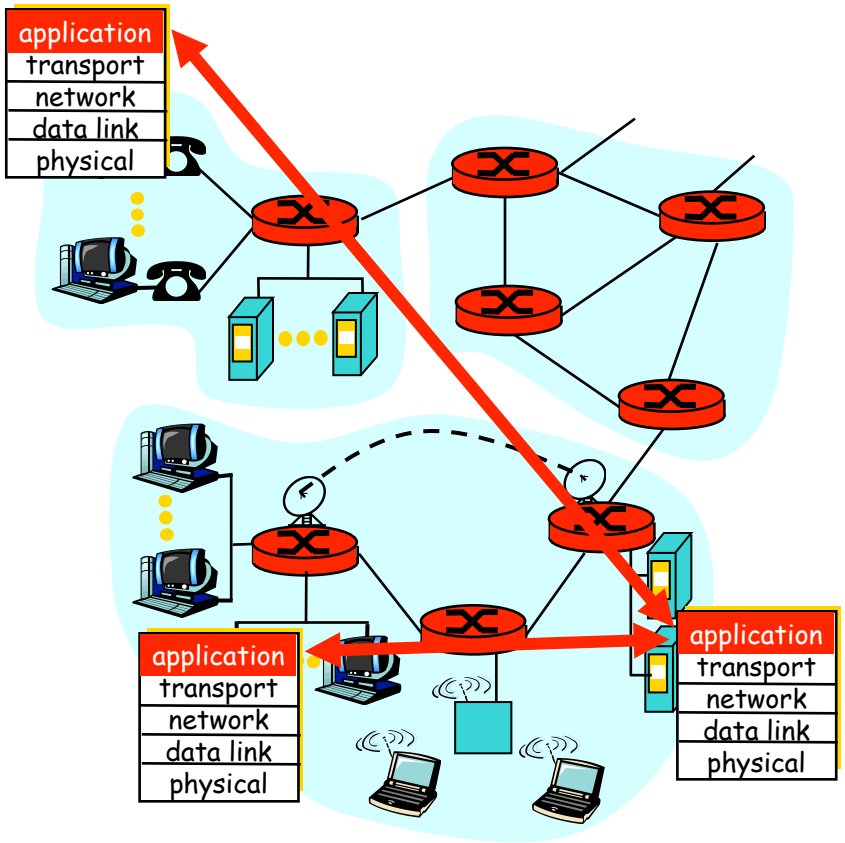
Chapter 7
Application Layer

第7章 应用层

- 概述
- DNS域名系统
- 电子邮件
- 万维网
- 流式音视频
- 内容分发

应用层概述（1）

- 应用程序：互相通信的分布的进程
  - 在网络主机上的用户空间运行
  - 互相交换消息
  - 比如email、ftp和web
- 应用层协议
  - 应用程序的一部分
  - 定义应用程序直接交换的信息以及相应的动作
  - 利用底层协议提供的服务



应用层概述（2）

- 应用层术语
  - 一个**进程**是运行于主机上的一个程序。
  - 在同一主机上的进程利用操作系统提供的IPC（interprocess communication）进行通信。
  - 在不同主机上运行的进程利用应用层协议进行通信
  - **用户代理**（user agent）是指用户和网络应用程序间的接口。比如web浏览器，流媒体播放器等

应用层概述（3）

- 应用程序编程接口API（application programming interface）
  - 定义应用程序和传输层之间的接口
  - socket: Internet API
    - 两个进程通过向socket写数据和读数据来通信
- Q: 进程如何指明它要与之通信的另一个进程
  - IP地址指明该进程所在的主机
  - 端口号指明该主机应该把收到的数据交给哪个当地进程

应用层概述（4）

- 应用程序需要什么样的传输服务
  - 数据丢失
    - 有的应用程序(比如音频)可以容忍一定程度的数据丢失
    - 有的应用程序(如文件传输)要求100%的可靠传输
  - 带宽
    - 有的程序(如多媒体)需要一定的带宽才能工作
    - 有的程序则使用它所能得到的全部带宽
  - 延迟
    - 有的程序要求低延迟，比如IP电话和交互游戏 .

应用程序所需的传输服务

应用	数据丢失	带宽	时间敏感
file transfer	no loss	elastic	no
e-mail	no loss	elastic	no
Web documents	no loss	elastic	no
real-time audio/video	loss-tolerant	audio: 5Kb-1Mb video:10Kb-5Mb	yes, 100’s msec
stored audio/video	loss-tolerant	same as above	yes, few secs
interactive games	loss-tolerant	few Kbps up	yes, 100’s msec
financial apps	no loss	elastic	yes and no

Internet传输协议提供的服务

TCP服务：

- 面向连接：用户端和服务器需要建立连接
- 接收和发送进程间的可靠传输
- 流量控制：发送方不会淹没接收方
- 拥塞控制：网络负载过高时限制发送方发送
- 不提供：延迟保证，最小带宽保证

UDP服务：

- 接收和发送进程间的不可靠传输
- 不提供：连接建立，可靠性、流量控制、拥塞控制和带宽保证

Internet应用和使用的相应协议

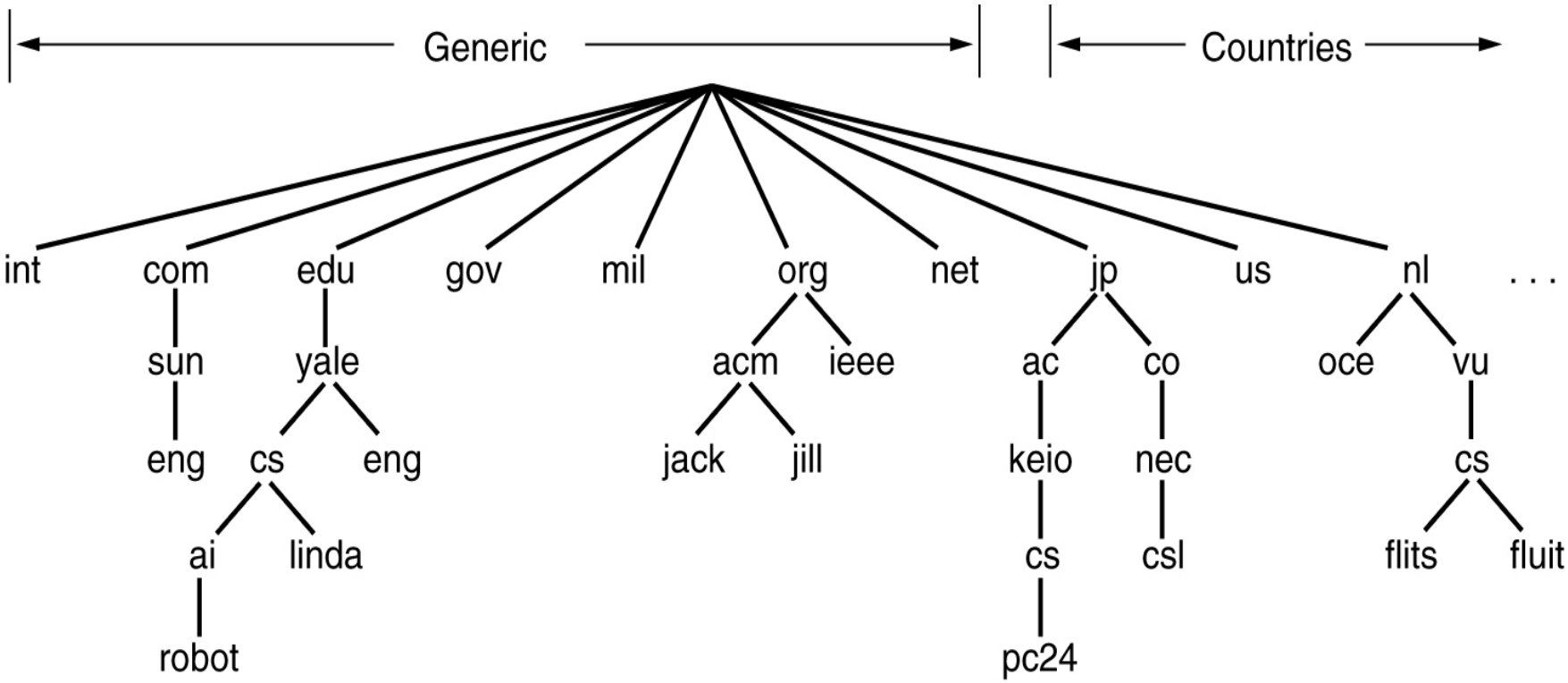
应用	应用层协议	底层传输层协议
e-mail	smtp [RFC 821]	TCP
remote terminal access	telnet [RFC 854]	TCP
Web	http [RFC 2068]	TCP
file transfer	ftp [RFC 959]	TCP
streaming multimedia	proprietary (e.g. RealNetworks)	TCP or UDP
remote file server	NSF	TCP or UDP
Internet telephony	proprietary (e.g., Vocaltec)	typically UDP

7.1 DNS域名系统

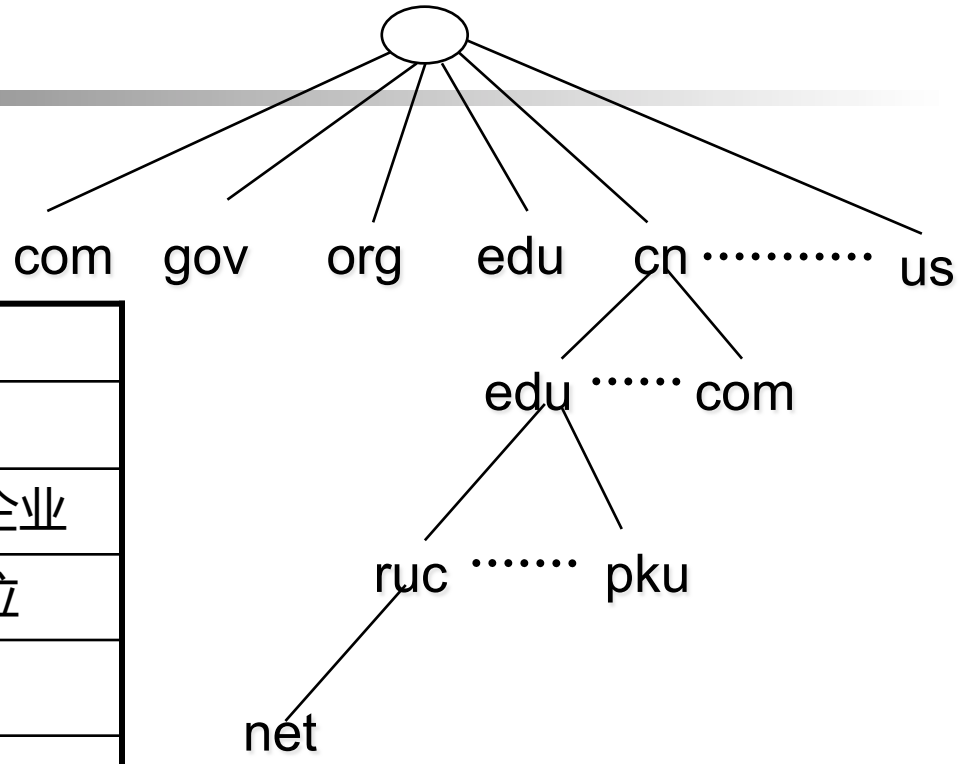
- DNS名字空间
- 域名资源记录
- 域名服务器

DNS名字空间

- 因特网名字与数字地址分配机构ICANN



DNS名字空间(2)



域名	意义	域名	意义
com	公司企业	firm	公司企业
net	网络服务机构	shop	销售公司企业
edu	教育机构	web	万维网单位
org	非赢利组织	arts	文化艺术
gov	美国政府部门	rec	娱乐消遣
mil	美国军事部门	info	信息服务
int	国际组织	nom	个人
cn	中国	jp	日本
us	美国	uk	英国

www.ruc.edu.cn
net.ruc.edu.cn

资源记录

- 资源记录格式
  - Domain_name, Time_to_live, Class, Type, Value
  - 域名
    记录所在域
  - 保留时间
  - 类别
  - 类型
  - 值

类 型	意 义	值
SOA	授权起始	本区域参数
A	主机IPv4地址	32bit
AAAA	主机IPv6地址	128bit
MX	邮件交换	优先级
NS	域名服务器	本域服务器名
CNAME	规范名	域名
PTR	指针	IP地址别名
SPF	发送者政策框架	邮件发送政策文本编码
SRV	服务	提供服务的主机
TXT	文本	任意描述文本

资源记录(2)

```
; Authoritative data for cs.vu.nl
cs.vu.nl.          86400   IN  SOA     star boss (952771,7200,7200,2419200,86400)
cs.vu.nl.          86400   IN  TXT     "Divisie Wiskunde en Informatica."
cs.vu.nl.          86400   IN  TXT     "Vrije Universiteit Amsterdam."
cs.vu.nl.          86400   IN  MX      1 zephyr.cs.vu.nl.
cs.vu.nl.          86400   IN  MX      2 top.cs.vu.nl.

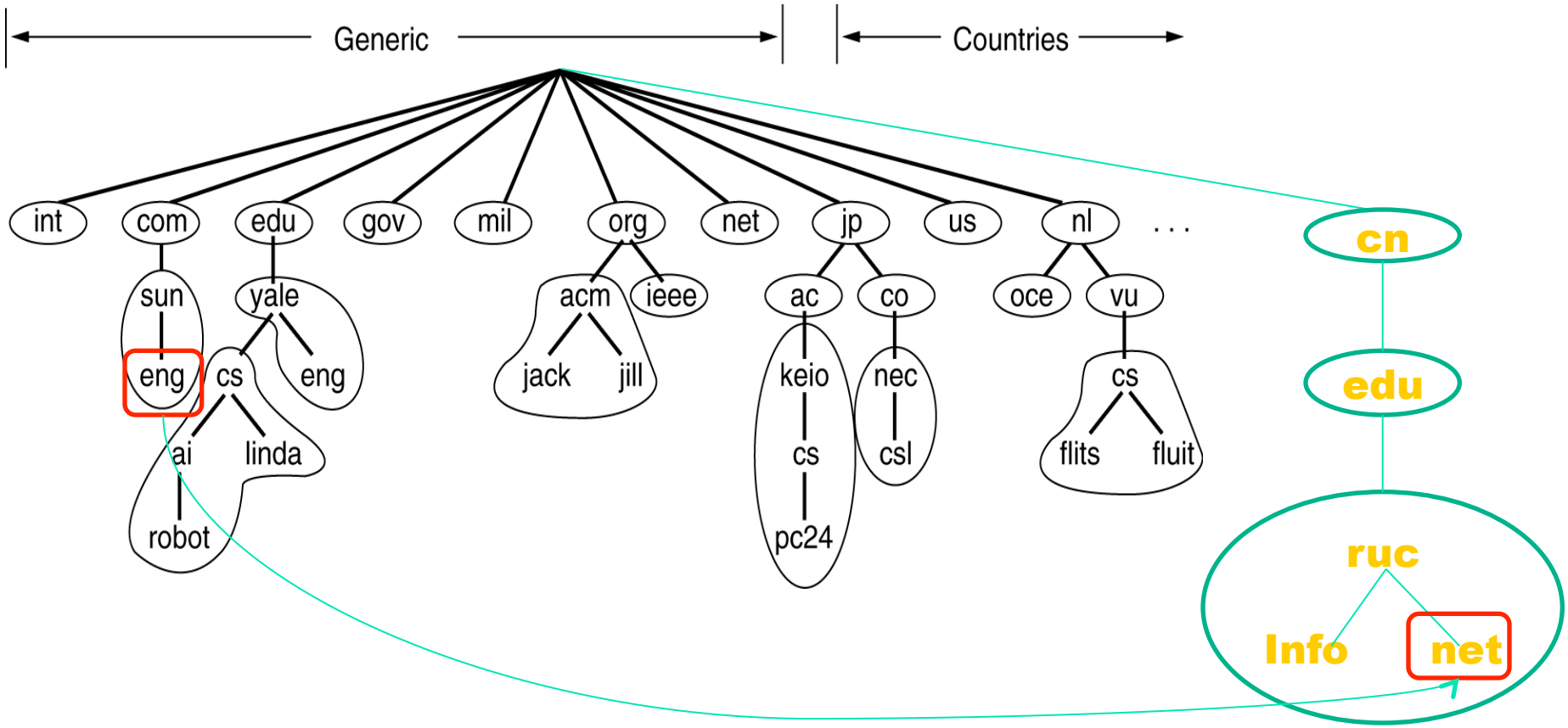
flits.cs.vu.nl.    86400   IN  HINFO   Sun Unix
flits.cs.vu.nl.    86400   IN  A       130.37.16.112
flits.cs.vu.nl.    86400   IN  A       192.31.231.165
flits.cs.vu.nl.    86400   IN  MX      1 flits.cs.vu.nl.
flits.cs.vu.nl.    86400   IN  MX      2 zephyr.cs.vu.nl.
flits.cs.vu.nl.    86400   IN  MX      3 top.cs.vu.nl.
www.cs.vu.nl.      86400   IN  CNAME   star.cs.vu.nl
ftp.cs.vu.nl.      86400   IN  CNAME   zephyr.cs.vu.nl

rowboat                    IN  A       130.37.56.201
                           IN  MX      1 rowboat
                           IN  MX      2 zephyr
                           IN  HINFO   Sun Unix

little-sister              IN  A       130.37.62.23
                           IN  HINFO   Mac MacOS

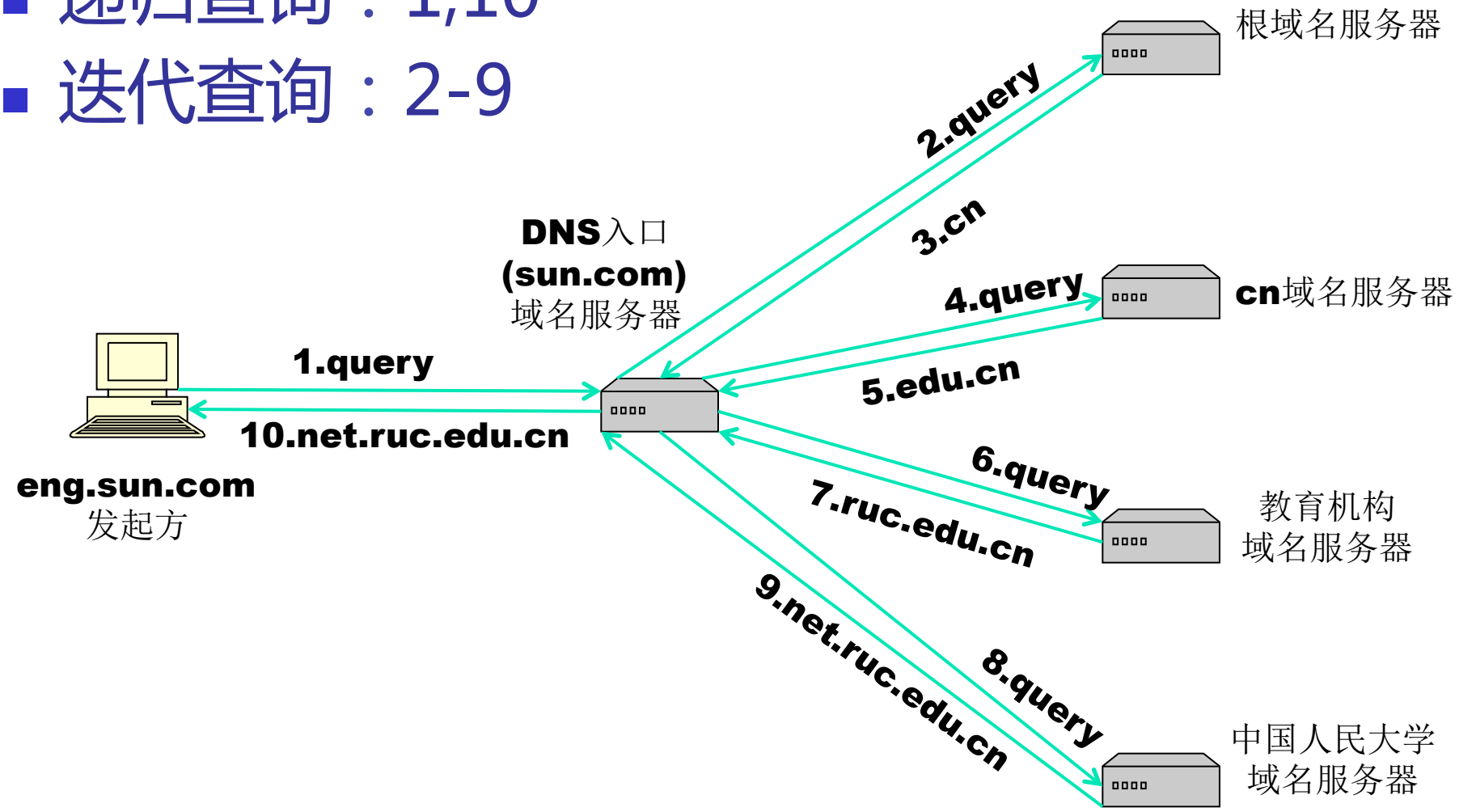
laserjet                   IN  A       192.31.231.216
                           IN  HINFO   "HP Laserjet IIISi" Proprietary
```

域名服务器


Generic
Countries
int
com
edu
gov
mil
org
net
jp
us
nl
. . .
sun
eng
yale
cs
eng
ai
linda
robot
acm
ieee
jack
jill
ac
co
keio
cs
pc24
nec
csl
oce
vu
cs
flits
fluit
cn
edu
ruc
Info
net

域名服务器(2)

- 递归查询：1,10
- 迭代查询：2-9

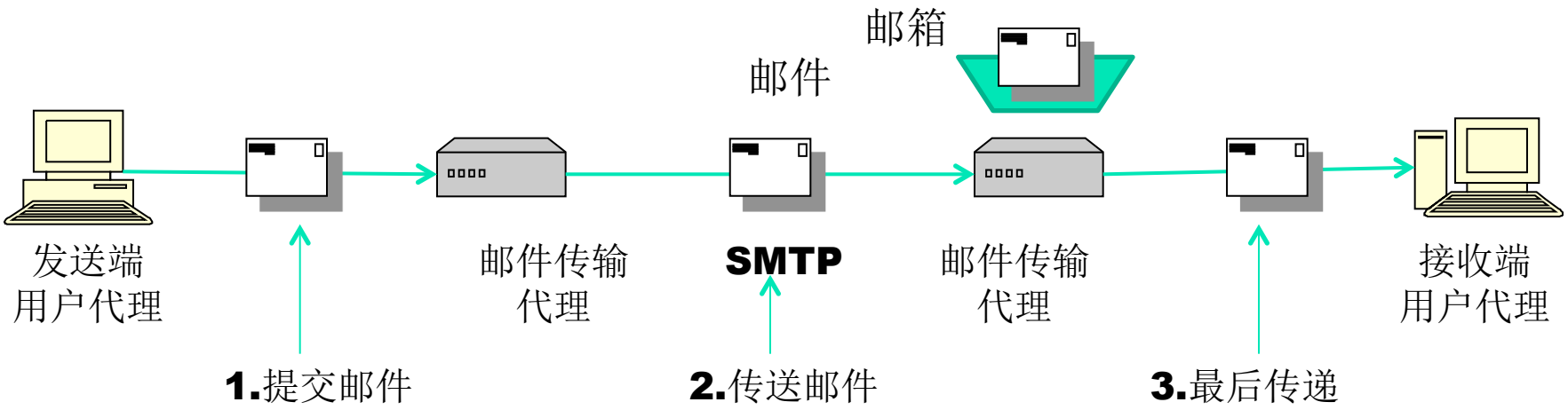


7.2 电子邮件

- 体系结构和服务
- 用户代理
- 邮件格式(RFC 5322)
- 邮件传送
- 最后传递

体系结构和服务

- 子系统：用户代理、传输代理
- 功能与协议：编写、显示、处理；简单邮件传输协议SMTP
- 其他：邮箱、收件人列表；信封、头部、主体



用户代理

邮件格式

- RFC 5322（822）头域

Header	Meaning
To:	E-mail address(es) of primary recipient(s)
Cc:	E-mail address(es) of secondary recipient(s)
Bcc:	E-mail address(es) for blind carbon copies
From:	Person or people who created the message
Sender:	E-mail address of the actual sender
Received:	Line added by each transfer agent along the route
Return-Path:	Can be used to identify a path back to the sender

Header	Meaning
Date:	The date and time the message was sent
Reply-To:	E-mail address to which replies should be sent
Message-Id:	Unique number for referencing this message later
In-Reply-To:	Message-Id of the message to which this is a reply
References:	Other relevant Message-Ids
Keywords:	User-chosen keywords
Subject:	Short summary of the message for the one-line display

邮件传送

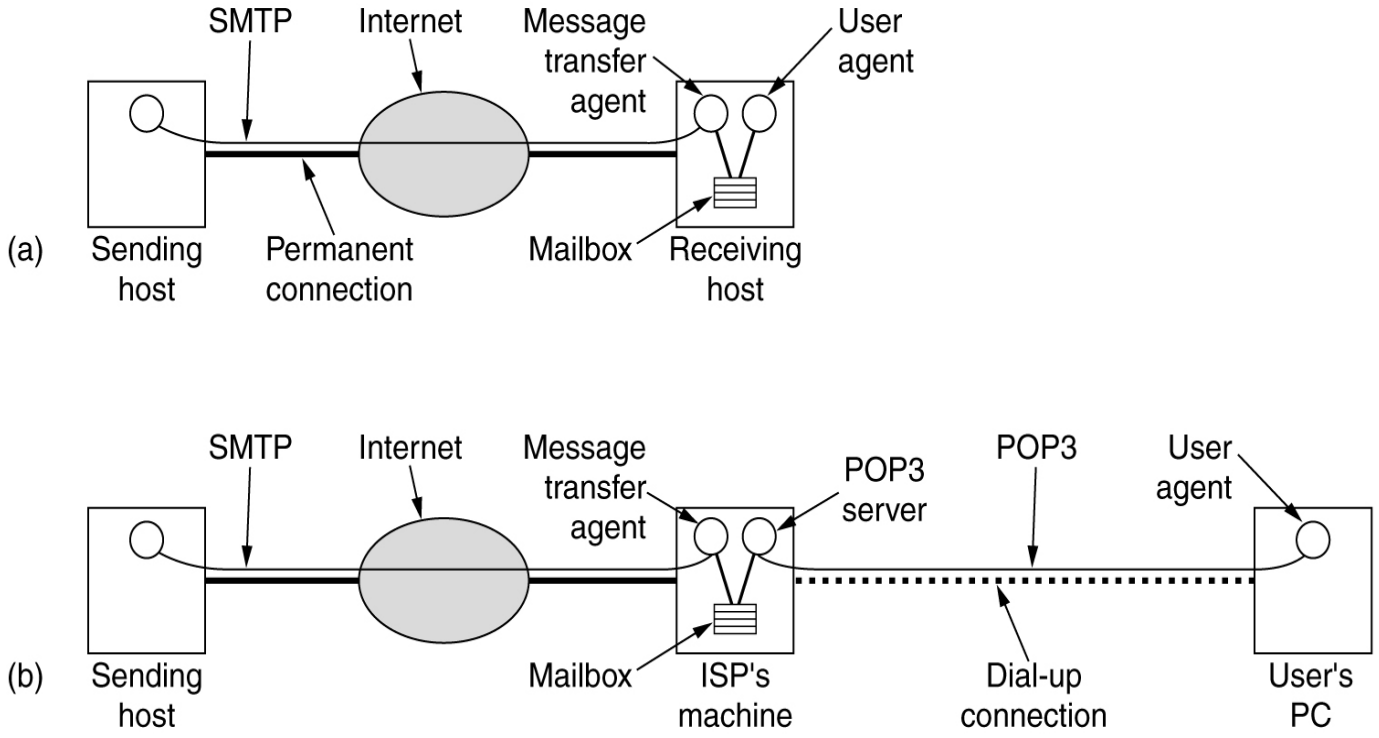
- SMTP
  - 简单邮件传输协议
  - 端口25

elinore@abc.com 发送邮件给*carolyn@xyz.com*

```
                S: 220 xyz.com SMTP service ready
C: HELO abcd.com
                S: 250 xyz.com says hello to abcd.com
C: MAIL FROM: <elinor@abcd.com>
                S: 250 sender ok
C: RCPT TO: <carolyn@xyz.com>
                S: 250 recipient ok
C: DATA
                S: 354 Send mail; end with "." on a line by itself
C: From: elinor@abcd.com
C: To: carolyn@xyz.com
C: MIME-Version: 1.0
C: Message-Id: <0704760941.AA00747@abcd.com>
C: Content-Type: multipart/alternative; boundary=qwertyuiopasdfghjklzxcvbnm
C: Subject: Earth orbits sun integral number of times
C:
C: This is the preamble. The user agent ignores it. Have a nice day.
C:
C: --qwertyuiopasdfghjklzxcvbnm
C: Content-Type: text/enriched
C:
C: Happy birthday to you
C: Happy birthday to you
C: Happy birthday dear <bold> Carolyn </bold>
C: Happy birthday to you
C:
C: --qwertyuiopasdfghjklzxcvbnm
C: Content-Type: message/external-body;
C:          access-type="anon-ftp";
C:          site="bicycle.abcd.com";
C:          directory="pub";
C:          name="birthday.snd"
C:
C: content-type: audio/basic
C: content-transfer-encoding: base64
C: --qwertyuiopasdfghjklzxcvbnm
C: .
                S: 250 message accepted
C: QUIT
                S: 221 xyz.com closing connection
```

最后的投递

- POP3——邮局协议
- IMAP——交互式电子邮件访问协议
- Webmail

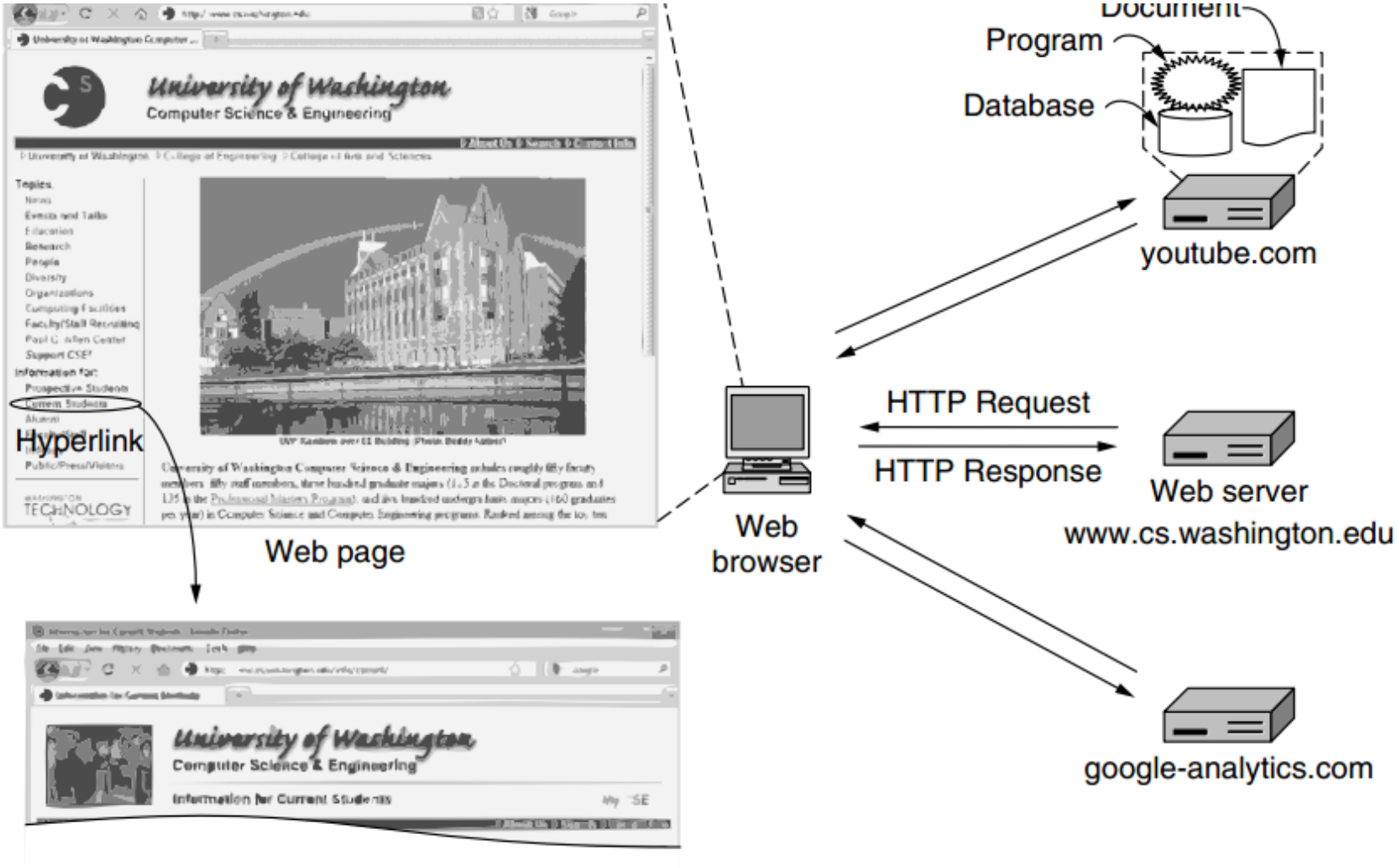


7.3 万维网

- 体系结构概述
- 静态Web页面
- 动态Web页面和Web应用
- HTTP——超文本传输协议
- 移动Web
- Web搜索

体系结构概述

- HTTP


University of Washington
Computer Science & Engineering
Hyperlink
Web page
Document
Program
Database
youtube.com
HTTP Request
HTTP Response
Web server
www.cs.washington.edu
Web browser
google-analytics.com

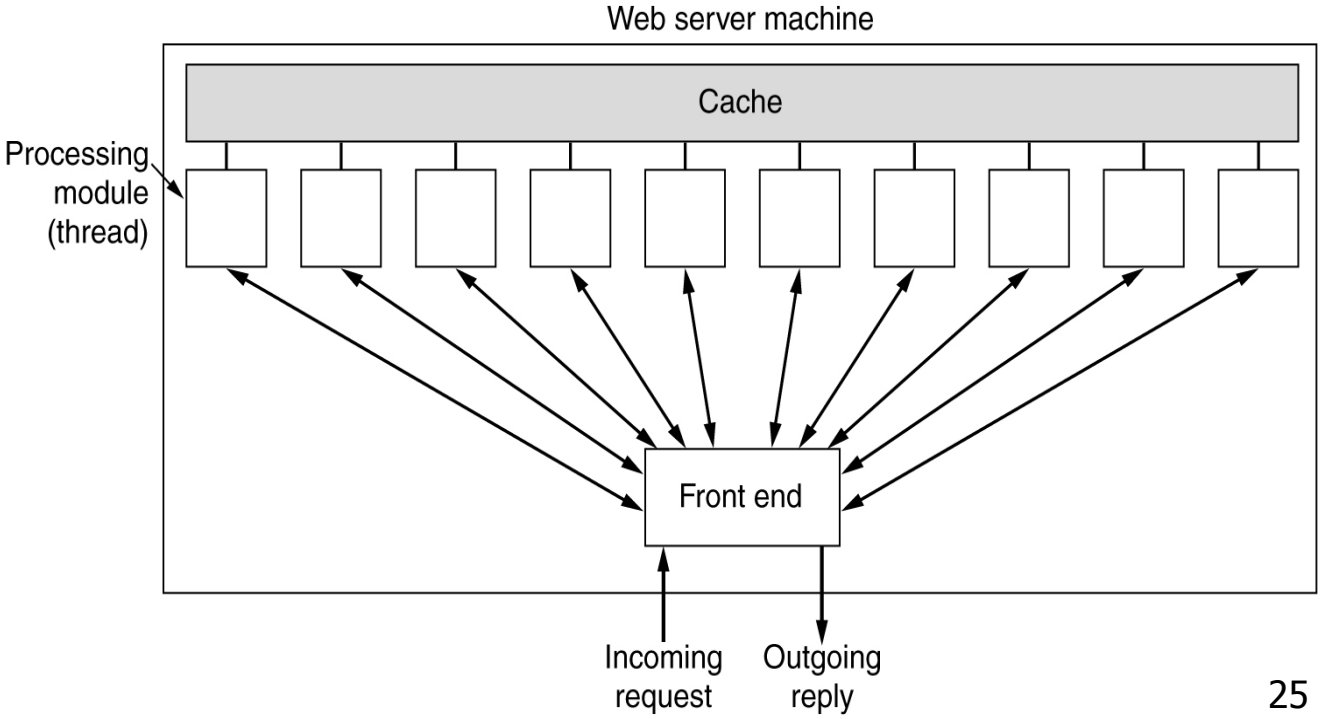
客户端

- 统一资源定位符URL
  - http://www.ruc.edu.cn:80/index.html
  - 202.112.126.221
- 超文本传输协议HTTP
- 浏览器、插件、辅助应用程序
- 多用途网际邮件扩充协议MIME (Multipurpose Internet Mail Extensions)

名　字	用　于	例　子
http	超文本	http://www.ee.uwa.edu/~rob/
https	安全的超文本	https://www.bank.com/accounts/
ftp	FTP	ftp://ftp.cs.vu.nl/pub/minix/README
file	本地文件	file:///usr/suzanne/prog.c
mailto	发送电子邮件	mailto:JohnUser@acm.org
rtsp	流式媒体	rtsp://youtube.com/montypython.mpg
sip	多媒体呼叫	sip:eve@adversary.com
about	浏览器信息	about:plugins

服务器端

- 监听TCP 80端口
- Web服务器
  - 内存缓存技术
  - 多线程模式



Cookie

- 内容（5个字段）

Domain	Path	Content	Expires	Secure
toms-casino.com	/	CustomerID=497793521	15-10-02 17:00	Yes
joes-store.com	/	Cart=1-00501;1-07031;2-13721	11-10-02 14:22	No
aportal.com	/	Prefs=Stk:SUNW+ORCL;Spt:Jets	31-12-10 23:59	No
sneaky.com	/	UserID=3627239101	31-12-12 23:59	No

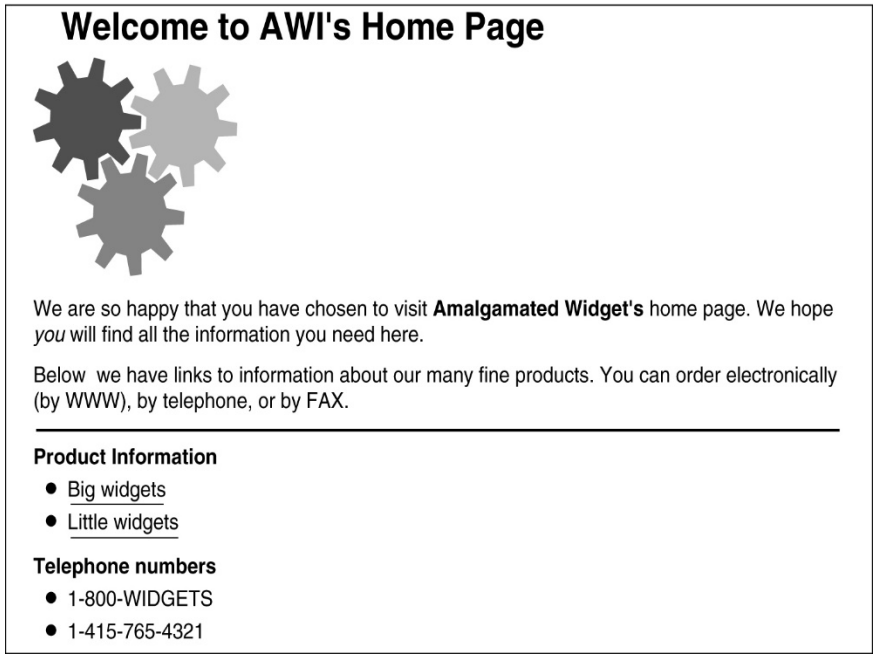
- 使用
  - 标识顾客
  - 商品列表（购物车）
  - 参数设置（Web门户网站）
  - 服务器访问记数

静态Web文档

- HTML——超文本标记语言

```
<html>
<head><title> AMALGAMATED  WIDGET,  INC. </title> </head>
<body> <h1> Welcome to AWI's Home Page</h1>
<img src="http://www.widget.com/images/logo.gif" ALT="AWI Logo"> <br>
We are so happy that you have chosen to visit <b> Amalgamated Widget's </b>
home page. We hope <i> you </i> will find all the information you need here.
<p>Below we have links to information about our many fine products.
You can order electronically (by WWW), by telephone, or by fax. </p>
<hr>
<h2> Product information  </h2>
<ul>
   <li> <a href="http://widget.com/products/big">  Big widgets </a>
   <li> <a href="http://widget.com/products/little">  Little widgets </a>
</ul>
<h2> Telephone numbers</h2>
<ul>
   <li> By telephone: 1-800-WIDGETS
   <li> By fax: 1-415-765-4321
</ul>
</body>
</html>
```

(a)



HTML——超文本标记语言

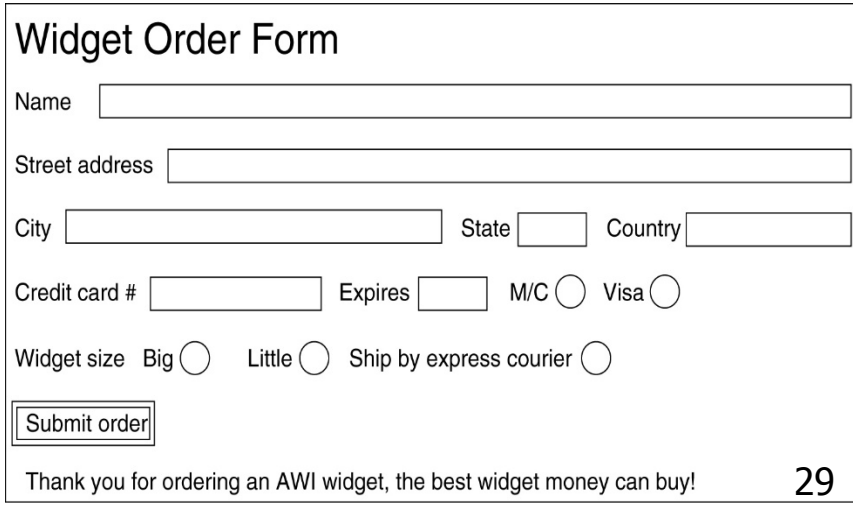
Tag	Description
<html> ... </html>	Declares the Web page to be written in HTML
<head> ... </head>	Delimits the page's head
<title> ... </title>	Defines the title (not displayed on the page)
<body> ... </body>	Delimits the page's body
<h *n*> ... </h*n*>	Delimits a level *n* heading
...	Set ... in boldface
...	Set ... in italics
...	Center ... on the page horizontally
...	Brackets an unordered (bulleted) li
...	Brackets a numbered list
	Starts a list item (there is no)
 	Forces a line break here
<p>	Starts a paragraph
<hr>	Inserts a Horizontal rule
<img src="...">	Displays an image here
 ... 	Defines a hyperlink

Item	HTML 1.0	HTML 2.0	HTML 3.0	HTML 4.0	HTML 5.0
Hyperlinks	x	x	x	x	x
Images	x	x	x	x	x
Lists	x	x	x	x	x
Active maps & images		x	x	x	x
Forms		x	x	x	x
Equations			x	x	x
Toolbars			x	x	x
Tables			x	x	x
Accessibility features				x	x
Object embedding				x	x
Style sheets				x	x
Scripting				x	x
Video and audio					x
Inline vector graphics					x
XML representation					x
Background threads					x
Browser storage					x
Drawing canvas					x

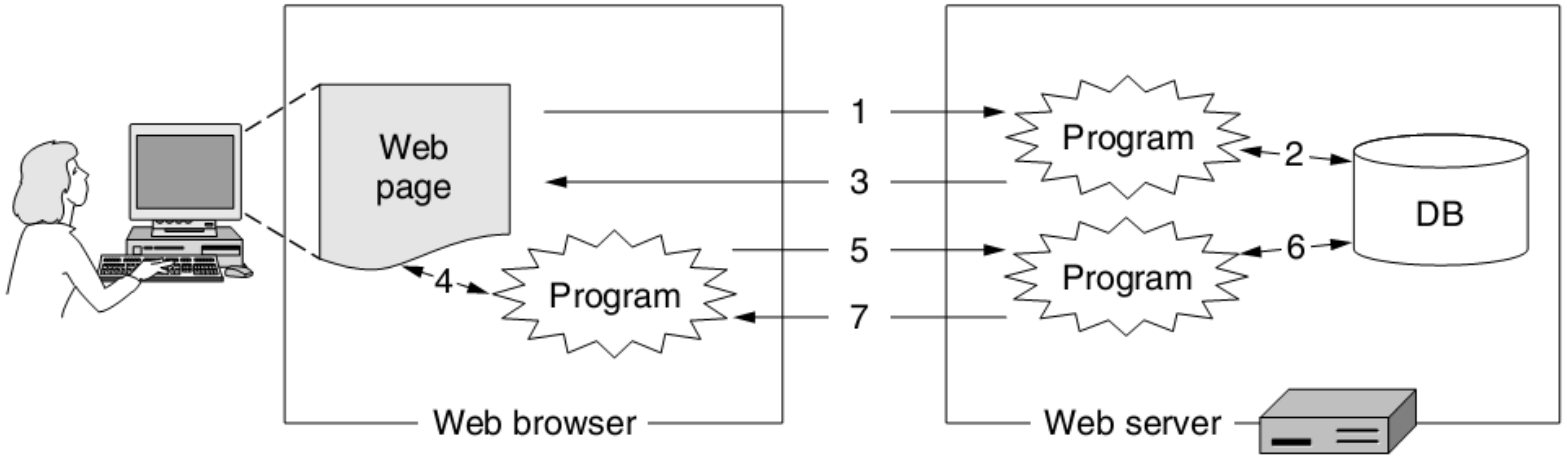
输入与表单

```
<html>
<head> <title> AWI CUSTOMER ORDERING FORM </title> </head>
<body>
<h1> Widget Order Form </h1>
<form ACTION="http://widget.com/cgi-bin/widgetorder" method=POST>
<p> Name <input name="customer" size=46> </p>
<p> Street Address <input name="address" size=40> </p>
<p> City <input name="city" size=20> State <input name="state" size =4>
Country <input name="country" size=10> </p>
<p> Credit card # <input name="cardno" size=10>
Expires <input name="expires" size=4>
M/C <input name="cc" type=radio value="mastercard">
VISA <input name="cc" type=radio value="visacard"> </p>
<p> Widget size Big <input name="product" type=radio value="expensive">
Little <input name="product" type=radio value="cheap">
Ship by express courier <input name="express" type=checkbox> </p>
<p><input type=submit value="submit order"> </p>
Thank you for ordering an AWI widget, the best widget money can buy!
</form>
</body>
</html>
```

(a)



动态Web页面与Web应用



- 服务器端动态Web页面生成
- 客户端动态Web页面生成

服务器端动态Web页面生成

- 公共网关接口CGI
  - 包含表单的Web页面
- 超文本预处理器
  - PHP脚本
- 其他
  - JSP
  - ASP

```
<html>
<body>
<form action="action.php" method="post">
<p> Please enter your name: <input type="text" name="name"> </p>
<p> Please enter your age: <input type="text" name="age"> </p>
<input type="submit">
</form>
</body>
</html>
```

(a)

```
<html>
<body>
<h1> Reply: </h1>
Hello <?php echo $name; ?>.
Prediction: next year you will be <?php echo $age + 1; ?>
</body>
</html>
```

(b)

```
<html>
<body>
<h1> Reply: </h1>
Hello Barbara.
Prediction: next year you will be 25
</body>
</html>
```

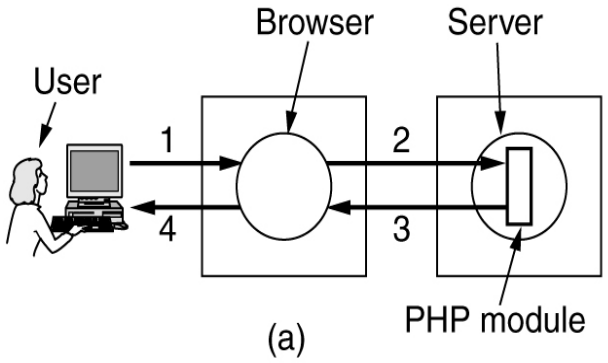
(c)

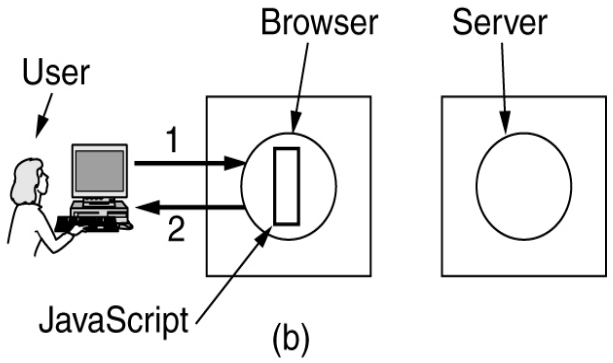
客户端动态网页生成

- 动态HTML
  - JavaScript
  - VBScript

```
<head>
<script language="javascript" type="text/javascript">
function response(test  form) {
   var person = test  form.name.value;
   var years = eval(test  form.age.value) + 1;
   document.open();
   document.writeln("<html> <body>");
   document.writeln("Hello " + person + ".<br>");
   document.writeln("Prediction: next year you will be " + years + ".");
   document.writeln("</body> </html>");
   document.close();
}
</script>
</head>

<body>
<form>
Please enter your name: <input type="text" name="name">
<p>
Please enter your age: <input type="text" name="age">
<p>
<input type="button" value="submit" onclick="response(this.form)">
</form>
</body>
</html>
```





HTTP——超文本传输协议

- 发展趋势
  - HTTP像一个传输协议
  - SOAP：运行在HTTP之上的XML RPC
- 连接

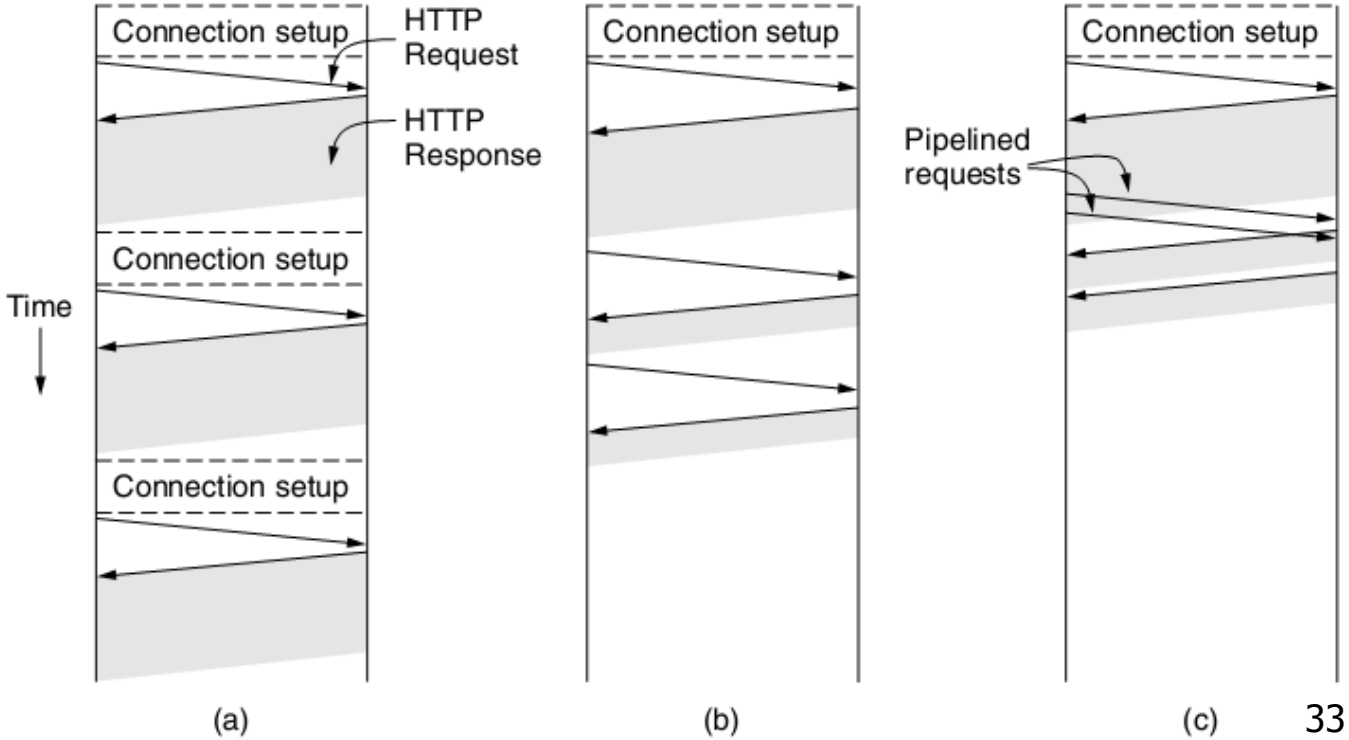
  HTTP1.1

  持续连接

  a.多个连接

  b.一个持续连接

  c.流水线式请求
- 方法

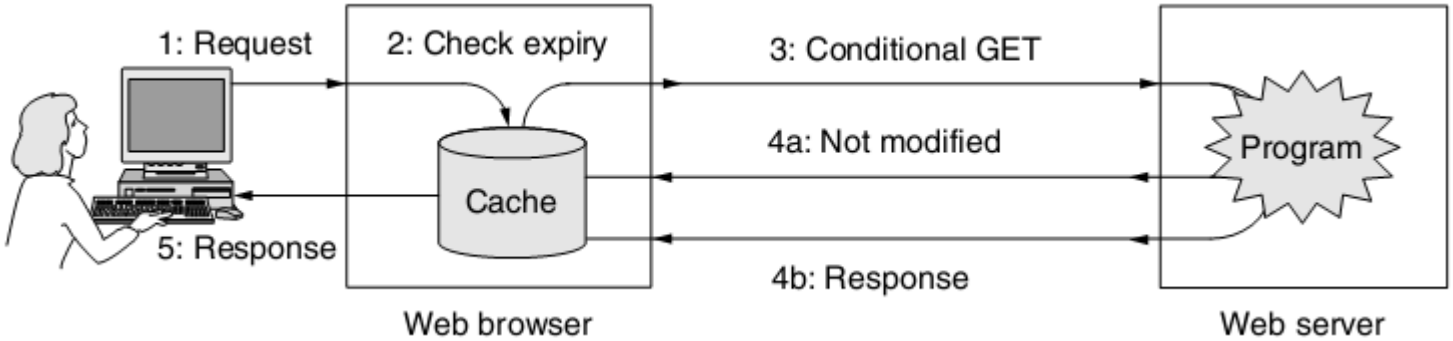


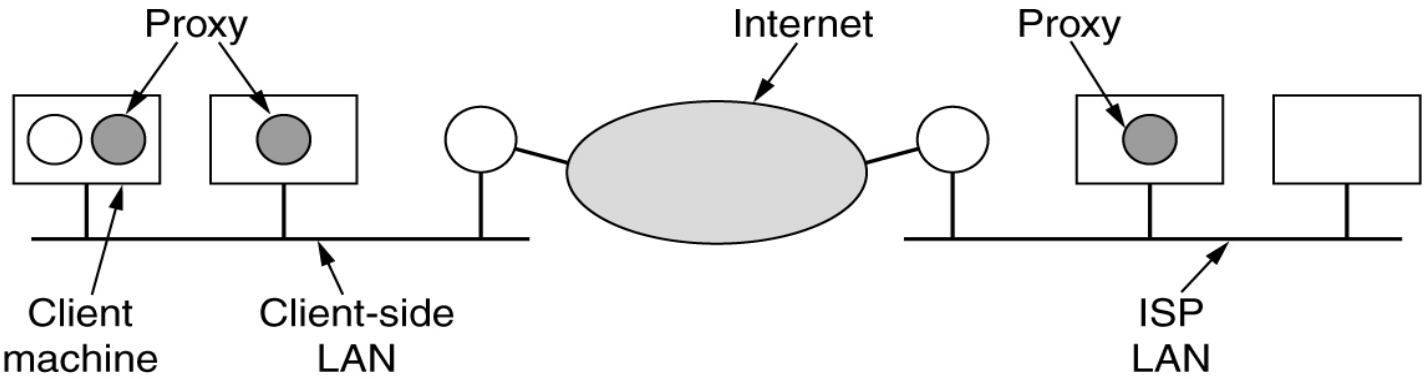
HTTP——超文本传输协议(2)

方　法	描　述
GET	请求读万维网页
HEAD	请求读万维网页头
PUT	请求存储万维网页
POST	附加一个命名资源(如万维网页)
DELETE	删除万维网页
TRACE	送回收到的请求
CONNECT	保留将来使用
OPTIONS	查询特定选项

HTTP——超文本传输协议(3)

- 消息头
- 缓存





- HTTP实验

移动Web

- WAP——无线应用协议
- 第二代无线Web
  - XHTML

XHTML	
WSP	HTTP
WTP	TLS
WTLS	TCP
WDP	IP
Bearer layer	Bearer layer
WAP 1.0 protocol stack	WAP 2.0 protocol stack

Module	Req.?	Function	Example tags
Structure	yes	Doc. structure	body, head, html, title
Text	yes	Information	br, code, dfn, em, h*n*, kbd, p, strong
Hypertext	yes	Hyperlinks	a
List	yes	Itemized lists	dl, dt, dd, ol, ul, li
Forms	No	Fill-in forms	form, input, label, option, textarea
Tables	No	Rectangular tables	caption, table, td, th, tr
Image	No	Pictures	img
Object	No	Applets, maps, etc.	object, param
Meta-information	No	Extra info	meta
Link	No	Similar to <a>	link
Base	No	URL starting point	base

Web搜索

- Web爬虫
  - 静态页面
  - 动态页面
    - 深层Web
  - 爬虫获取数据的处理
  - 索引与命名

作业

- 1，6，16，37，42